CME 307 / MS&E 311 / OIT 676: Optimization

Optimality conditions and convexity

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Management Science and Engineering
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Outline

Constrained and unconstrained optimization

Optimality conditions

Convex analysis

Convex optimization

A smooth optimization problem

constrained optimization: minimize $f(x)$ over $x \in C \subseteq \mathbb{R}^n$ // unconstrained case: $C = \mathbb{R}^n$.

running example: least squares. with $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, solve

$$\text{minimize} \quad f(x) = \frac{1}{2}\|Ax - b\|^2.$$

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when can local information certify a **global** minimizer?

Oracles

an optimization **oracle** tells us what we can evaluate at a point x:

- **zero-order:** $f(x)$
- **first-order:** $f(x)$ and $\nabla f(x)$
- **second-order:** $f(x)$, $\nabla f(x)$, and $\nabla^2 f(x)$

the oracle model covers both explicit formulas and systems we can only query. it also lets us count the information an algorithm uses.

source: Nesterov 2004 "Introductory Lectures on Convex Optimization"

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Solution of an optimization problem

$$\text{minimize} \quad f(x)$$

for $f : \mathcal{D} \to \mathbb{R}$. $x^\star$ is a

- **global minimizer** if $f(x) \geq f(x^\star)$ for all $x \in \mathcal{D}$.
- **local minimizer** if there is a neighborhood $\mathcal{N}$ around $x^\star$ so that $f(x) \geq f(x^\star)$ for all $x \in \mathcal{N}$.
- **isolated local minimizer** if the neighborhood $\mathcal{N}$ contains no other local minimizers.
- **unique minimizer** if it is the only global minimizer.

Solution of an optimization problem

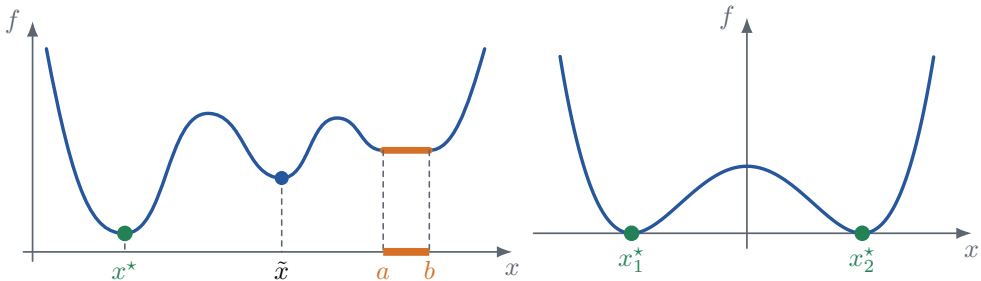
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pictures!

Local and global minimizers: pictures



left: $x^\star$ is the unique global minimizer; $\tilde{x}$ is an isolated local minimizer; every point of $[a, b]$ is a local minimizer, but none is isolated. right: two global minimizers, so neither is unique.

First order optimality condition

Theorem

If $x^\star \in \mathbb{R}^n$ is a local minimizer of a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, then $\nabla f(x^\star) = 0$.

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proof: suppose by contradiction that $\nabla f(x^\star) \neq 0$. consider points of the form $x_\alpha = x^\star - \alpha \nabla f(x^\star)$ for $\alpha > 0$. by definition of the gradient,

$$\lim_{\alpha \to 0} \frac{f(x_\alpha) - f(x^\star)}{\alpha} = -\nabla f(x^\star)^\top \nabla f(x^\star) = -\|\nabla f(x^\star)\|^2 < 0$$

so for any sufficiently small $\alpha > 0$, we have $f(x_\alpha) < f(x^\star)$, which contradicts the fact that $x^\star$ is a local minimizer.

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Definition

$x^\star \in \mathbb{R}^n$ is a **stationary point** of a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ if $\nabla f(x^\star) = 0$.

is a stationary point always a local minimizer?

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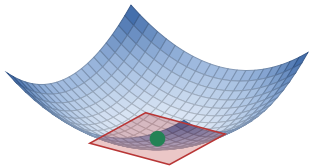
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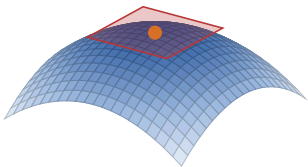
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is a stationary point always a local minimizer? no! saddle points, local maximizers.

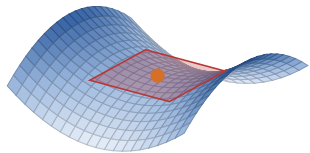
Stationary points



$x_1^2 + x_2^2$



$-x_1^2 - x_2^2$



$x_1^2 - x_2^2$

each function has $\nabla f(0) = 0$ (horizontal tangent plane at the origin): a minimizer, a maximizer, and a saddle point.

Least squares: a stationary point

for $f(x) = \frac{1}{2}\|Ax - b\|^2$,

$$\nabla f(x) = A^T(Ax - b), \qquad \nabla^2 f(x) = A^T A.$$

Least squares: a stationary point

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any solution $x^\star$ of the normal equations $A^T A x^\star = A^T b$ is stationary. does this make it a global minimizer?

we will answer this after introducing convexity.

Second order optimality condition

Theorem

If $x^\star \in \mathbb{R}^n$ is a local minimizer of a twice differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, then $\nabla^2 f(x^\star) \succeq 0$.

Second order optimality condition

Theorem

If $x^\star \in \mathbb{R}^n$ is a local minimizer of a twice differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, then $\nabla^2 f(x^\star) \succeq 0$.

proof: since $\nabla f(x^\star) = 0$, for any direction v the expansion is

$$f(x^\star + \alpha v) = f(x^\star) + \frac{\alpha^2}{2} v^T \nabla^2 f(x^\star) v + o(\alpha^2).$$

if $\nabla^2 f(x^\star)$ is not psd, some v has $v^T \nabla^2 f(x^\star) v < 0$. then $f(x^\star + \alpha v) < f(x^\star)$ for sufficiently small $\alpha > 0$, contradicting local minimality.

Symmetric positive semidefinite matrices

Definition

a symmetric matrix $Q \in \mathbb{R}^{n \times n}$ is **positive semidefinite** (psd) if $x^T Q x \geq 0$ for all $x \in \mathbb{R}^n$.

for symmetric $Q \in \mathbb{R}^{n \times n}$, these are equivalent:

$$Q \in \mathbf{S}_+^n \iff Q \succeq 0 \iff \lambda_{\min}(Q) \geq 0 \iff v^T Q v \geq 0 \quad \forall v \in \mathbb{R}^n$$

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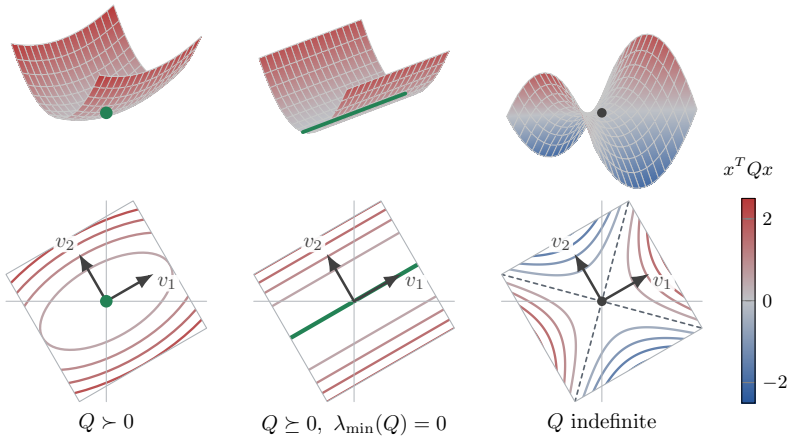
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$Q \in \mathbf{S}_{++}^n$ is **symmetric positive definite** (spd) ($Q \succ 0$) if $x^T Q x > 0$ for all $x \neq 0$.

Level sets of quadratic forms



top: the graph of $x^T Qx$; bottom: its level sets, in the same colors. eigenvalues $(\lambda_1, \lambda_2) = (0.5, 2)$, $(0, 2)$, $(2, -2)$ along v_1, v_2; green: minimizers.

Necessary and sufficient conditions

for unconstrained, smooth optimization: a local minimum is **strict** if $f(x) > f(x^\star)$ for every sufficiently close $x \neq x^\star$.

- local minimum $\implies \nabla f(x^\star) = 0$ and $\nabla^2 f(x^\star) \succeq 0$
- $\nabla f(x^\star) = 0$ and $\nabla^2 f(x^\star) \succ 0 \implies$ strict local minimum

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the necessary conditions alone do not suffice: $f(x) = x^3$ has $f'(0) = f''(0) = 0$, but 0 is not a local minimum.

Necessary and sufficient conditions

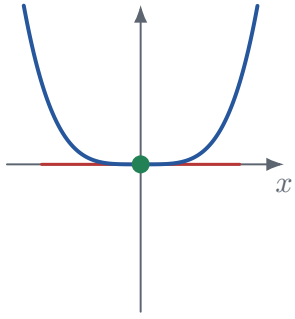
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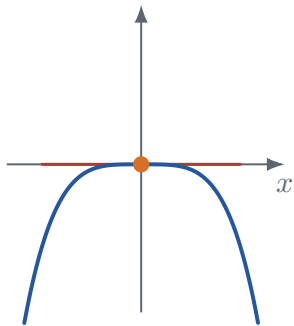
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convexity will give us a condition for **global** optimality.

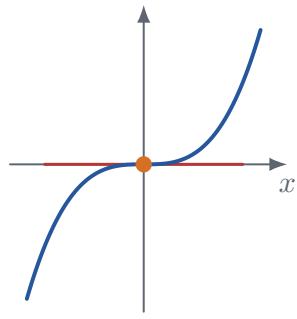
When second-order conditions are inconclusive



x^4



$-x^4$



x^3

all three have $f'(0) = 0$ and $f''(0) = 0$, so the second-order test is silent: 0 is a minimizer of x^4, a maximizer of $-x^4$, and neither for x^3.

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Convex sets

Definition

A set $S \subseteq \mathbb{R}^n$ is convex if it contains every chord: for all $\theta \in [0,1]$, w, $v \in S$,

$$\theta w + (1-\theta)v \in S$$

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Q: Which of these are convex?
ellipsoid, crescent moon, ...

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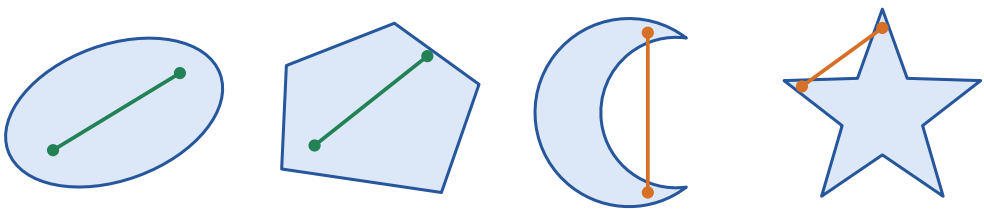
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Definition

The **relative interior** of S is its interior within its affine hull.

Convex or not?



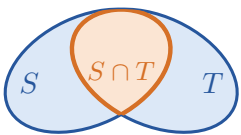
a single chord that leaves the set (orange) certifies that the set is not convex.

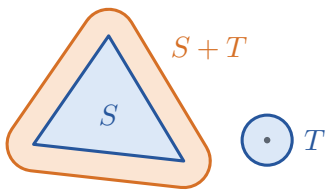
Operations that preserve convexity

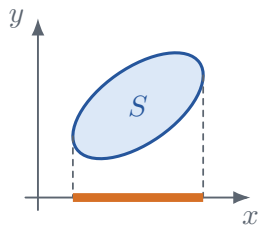
if $S \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}^n$ are convex, then so are:

- intersection: $S \cap T$
- sum: $S + T = \{s + t \mid s \in S,\ t \in T\}$
- projection: $\{x : (x, y) \in S\}$

Operations that preserve convexity: pictures







intersection $S \cap T$; sum $S + T$ of a triangle and a disc; projection of S onto the x-axis.

Convex functions

define the **extended real numbers** as $\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ and $f : \mathbb{R}^n \to \overline{\mathbb{R}}$, with **dom** $f = \{x : f(x) < +\infty\}$.

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- **Chords.** the function never lies above its chord: $\forall \theta \in (0,1)$, $w, v \in \mathbb{R}^n$

$$f(\theta w + (1-\theta)v) \leq \theta f(w) + (1-\theta) f(v)$$

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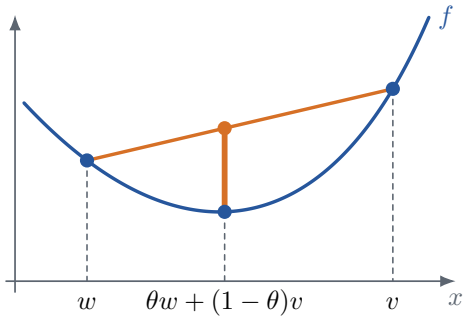
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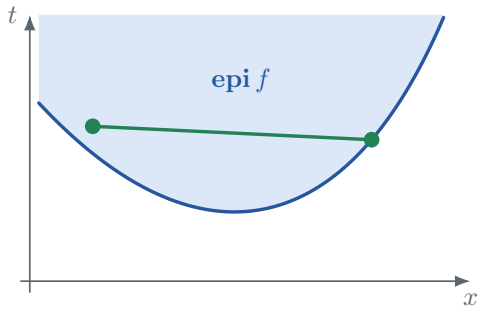
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- **Epigraph.** **epi**$(f) = \{(x,t) : \ t \geq f(x)\}$ is convex

Chords and epigraphs





left: the chord lies above the graph at $\theta w + (1-\theta)v$. right: the epigraph contains the chord between any two of its points.

Convexity and derivatives

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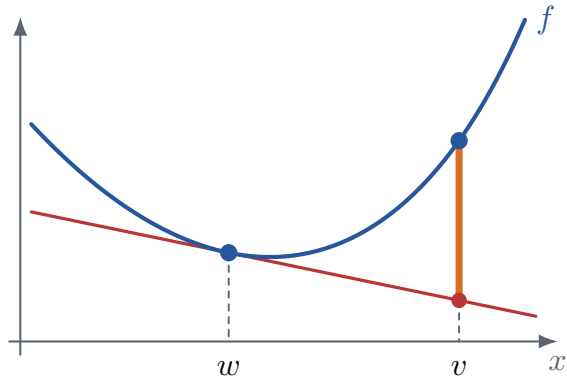
- **First order condition.** if f is differentiable,

$$f(v) \geq f(w) + \nabla f(w)^\top (v - w), \qquad \forall w, v \in \mathbf{dom}\, f$$

- **Second order condition.** if f is twice differentiable, its Hessian is always psd:

$$\lambda_{\min}(\nabla^2 f(x)) \geq 0, \qquad \forall x \in \mathbf{dom}\, f$$

First-order condition: picture



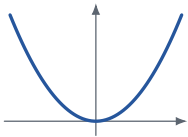
the tangent at w is a global underestimator of f: the gap at v is $f(v) - f(w) - \nabla f(w)^\top (v - w) \geq 0$.

Convexity examples

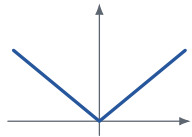
Q: Which of these functions are convex?

- quadratic function $f(x) = x^2$ for $x \in \mathbb{R}$
- absolute value function $f(x) = |x|$ for $x \in \mathbb{R}$
- quadratic function $f(x) = x^T Ax$, $x \in \mathbb{R}^n$, $A \succeq 0$
- quadratic function $f(x) = x^T Ax$, A indefinite
- rollercoaster function (cubic) $f(x) = (x-1)(x-3)(x-5)$
- hyperbolic function $f(x) = 1/x$ for $x > 0$
- jump function $f(x) = 1$ if $x \geq 0$, $f(x) = 0$ otherwise
- jump to infinity function $f(x) = 1$ if $x \in [-1, 1]$, $f(x) = \infty$ otherwise

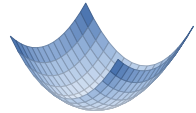
Convexity examples: answers



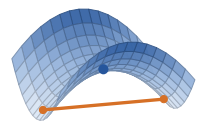
x^2



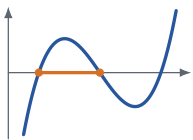
$|x|$



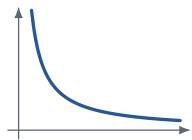
$x^\top Ax,\ A \succeq 0$



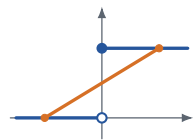
$x^\top Ax,\ A$ indefinite



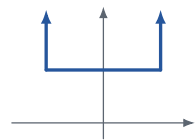
$(x-1)(x-3)(x-5)$



$1/x,\ x > 0$



$\mathbf{1}\{x \geq 0\}$



1 on $[-1, 1]$, $+\infty$ off

an orange chord that passes below the graph shows the function is not convex.

Operations that preserve convexity

if $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$ are convex, then so are:

- cf for $c \geq 0$
- $f(Ax + b)$ for $A \in \mathbb{R}^{n \times m}$, $b \in \mathbb{R}^n$
- $f + g$
- $\max\{f, g\}$

Operations that preserve convexity

if $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$ are convex, then so are:

- cf for $c \geq 0$
- $f(Ax + b)$ for $A \in \mathbb{R}^{n \times m}$, $b \in \mathbb{R}^n$
- $f + g$
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Q: Pick one and assume f and g are twice-differentiable. What is the easiest way to prove convexity?
a useful scalar composition rule:

$f \circ g(x) = f(g(x))$ is convex if g is convex and f is convex and nondecreasing

since

$$(f \circ g)''(x) = f''(g(x))(g'(x))^2 + f'(g(x))g''(x)$$

Jensen's inequality

Jensen's inequality generalizes the chord condition to a distribution of points:

Theorem

If $f : \mathbb{R}^n \to \mathbb{R}$ is convex and X is a random variable, then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

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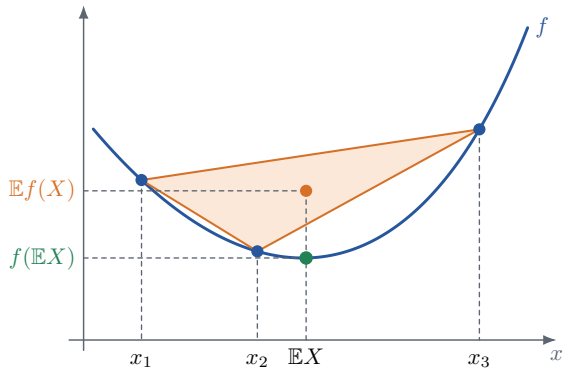
If $f : \mathbb{R}^n \to \mathbb{R}$ is convex and X is a random variable, then

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example: for positive probability vectors p, q, apply Jensen to $-\log$:

$$\begin{aligned} D_{\mathrm{KL}}(p\|q) &= \sum_i p_i \big[-\log(q_i/p_i)\big] \\ &\geq -\log\left(\sum_i p_i \frac{q_i}{p_i}\right) = -\log\left(\sum_i q_i\right) = 0. \end{aligned}$$

Jensen's inequality: picture



X takes values x_1, x_2, x_3 with probabilities p_1, p_2, p_3. the point $(\mathbb{E}X, \mathbb{E}f(X))$ is a convex combination of the points $(x_i, f(x_i))$, so it lies in their convex hull, which lies in **epi** f.

Sublevel set

Definition

The **sublevel set** of a function $f : \mathbb{R}^n \to \mathbb{R}$ at level t is

$$S_t = \{x \in \mathbb{R}^n \mid f(x) \leq t\}$$

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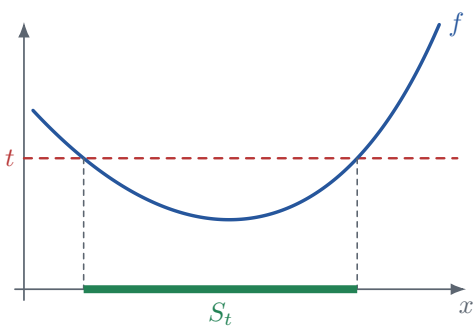
A convex function $f : \mathbb{R}^n \to \mathbb{R}$ has convex sublevel sets.

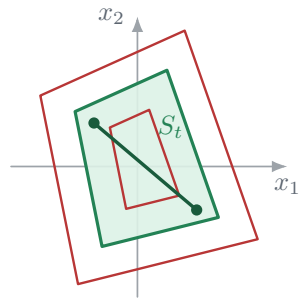
proof: Jensen's inequality. if $x, y \in S_t$, then for $\theta \in [0, 1]$,

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y) \leq \theta t + (1-\theta)t = t$$

so $\theta x + (1-\theta)y \in S_t$.

Sublevel sets: pictures





left: S_t is an interval. right: nested sublevel sets of a piecewise-linear convex function $f(x) = \max_i(a_i^T x + b_i)$, with S_t shaded.

Quasiconvexity

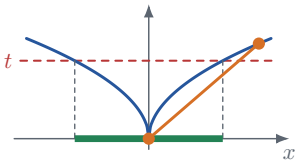
converse is not true: a function can have all sublevel sets convex, and still be non-convex.

Definition

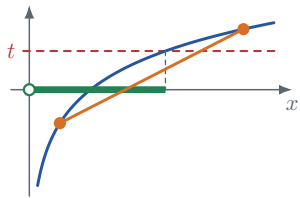
A function $f : \mathbb{R}^n \to \mathbb{R}$ is **quasiconvex** if its sublevel sets are convex.

examples of functions that are quasiconvex but not convex?

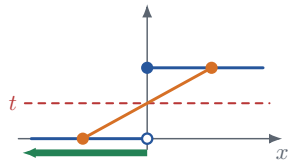
Quasiconvex but not convex



$\sqrt{|x|}$



$\log x,\ x > 0$



$\mathbf{1}\{x \geq 0\}$

each sublevel set S_t (green) is an interval, but a chord (orange) passes below the graph.

Supporting hyperplane

Definition

A **supporting hyperplane** to a set $S \subseteq \mathbb{R}^n$ at a point $x \in \overline{S}$ is a hyperplane that touches $\overline{S}$ at x and lies entirely on one side of S:

$$H = \{y \in \mathbb{R}^n \mid a^\top y = b\} \text{ supports } S \text{ at } x \text{ if } \quad \begin{array}{lll} a^\top x & = b & \\ a^\top y & \geq b & \forall y \in S \end{array}$$

Supporting hyperplane

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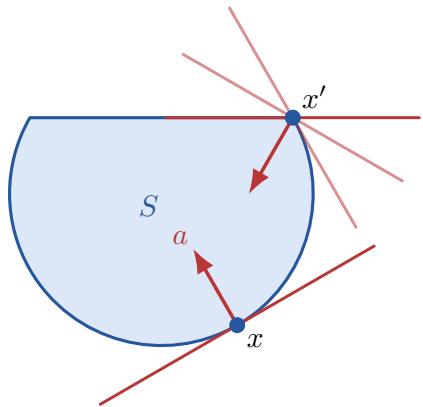
A **supporting hyperplane** to a set $S \subseteq \mathbb{R}^n$ at a point $x \in \overline{S}$ is a hyperplane that touches $\overline{S}$ at x and lies entirely on one side of S:

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Theorem (Supporting hyperplane)

Any nonempty convex set has a supporting hyperplane at every boundary point.

Supporting hyperplanes: picture



at a smooth boundary point x the supporting hyperplane is unique, with normal a pointing into S; at a corner x' there are many.

Supporting hyperplane condition for convexity

Theorem (Partial converse)

If a closed set with nonempty interior has a supporting hyperplane at every boundary point, then it is convex.

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*A function $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ is convex $\iff$ for all $x \in$ **relint dom** f, the epigraph of f has a supporting hyperplane at $(x, f(x))$: for some $g \in \mathbb{R}^n$,*

$$f(y) \geq f(x) + g^\top (y - x) \quad \forall y \in \mathbb{R}^n$$

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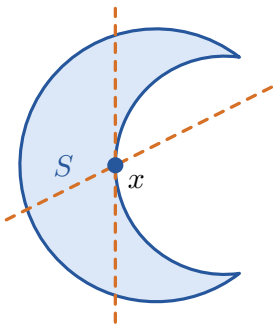
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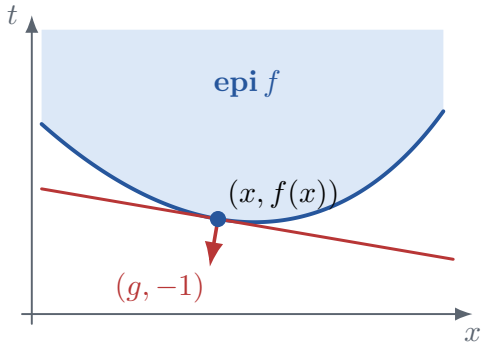
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Definition

A vector $g \in \mathbb{R}^n$ is a **subgradient** of $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ at $x \in$ **dom** f if $f(y) \geq f(x) + g^\top (y - x)$ for all $y \in \mathbb{R}^n$.

Supporting hyperplanes and convexity: pictures





left: at x, every line through x has points of S on both sides, so S is not convex.
right: a supporting hyperplane to **epi** f at $(x, f(x))$, with outward normal $(g, -1)$.

Example: subgradients

let $f = \max\{f_1, f_2\}$, where f_1, f_2 are convex and differentiable. **Q:** What is $\partial f(x)$ when one function is larger? What happens at a tie?

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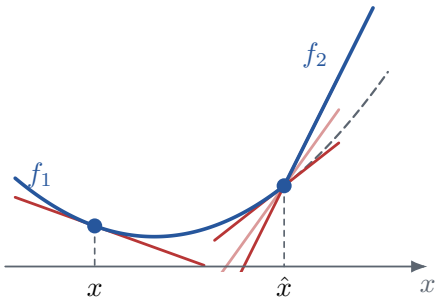
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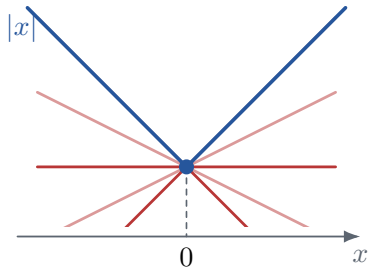
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example: $|x| = \max\{x, -x\}$, so $\partial|0| = [-1, 1]$.

Subgradients: pictures





left: $f = \max\{f_1, f_2\}$. at x, only f_1 is active and $\partial f(x) = \{\nabla f_1(x)\}$; at the tie $\hat{x}$, every slope between $\nabla f_1(\hat{x})$ and $\nabla f_2(\hat{x})$ gives a supporting line. right: $\partial|x|(0) = [-1, 1]$.

Subdifferential

set of all subgradients of f at x is called the **subdifferential** $\partial f(x)$

$$\partial f(x) = \{g : f(y) \geq f(x) + g^T(y - x) \quad \forall y\}$$

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if f is convex,

- $\partial f(x)$ is nonempty, for $x \in$ **relint dom** f
- $\partial f(x) = \{\nabla f(x)\}$, if f is differentiable at $x \in \text{int}\,\textbf{dom}\, f$
- if $\partial f(x) = \{g\}$, then f is differentiable at $x \in \text{int}\,\textbf{dom}\, f$ and $g = \nabla f(x)$

Global optimality with subgradients

for a convex function $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ and $x^\star \in \mathbf{dom}\, f$,

$$x^\star \text{ is a global minimizer} \quad \iff \quad 0 \in \partial f(x^\star).$$

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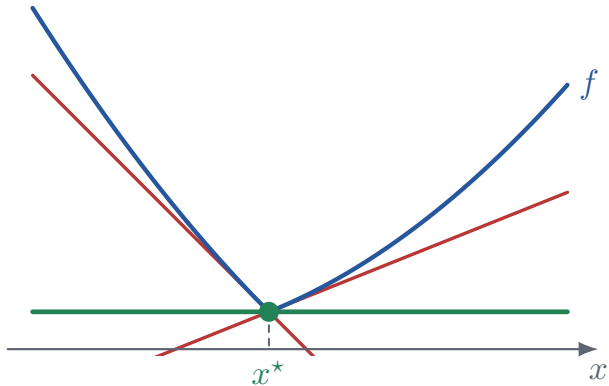
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for $f(x) = |x|$, we have $\partial f(0) = [-1, 1]$. since $0 \in \partial f(0)$, $x^\star = 0$ is globally optimal.

Global optimality with subgradients: picture



f is not differentiable at $x^\star$, but $0 \in \partial f(x^\star)$: the horizontal line (green) supports **epi** f, so $f(y) \geq f(x^\star)$ for all y.

Outline

Constrained and unconstrained optimization

Optimality conditions

Convex analysis

Convex optimization

Convex optimization

an optimization problem is convex if:

- **Geometrically:** the feasible set and the epigraph of the objective are convex

for example, a nonlinear minimization is convex if the objective and inequality constraints are convex functions, and the equality constraints are affine

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq b_i, \quad i = 1, \ldots, m_1 \\ & Ax = b_2 \\ \text{variable} & x \in \mathbb{R}^n \end{array}$$

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concave functions:

- a function f is concave if $-f$ is convex
- concave maximization $\implies$ a convex optimization problem

Why care about convex optimization?

- local optimality $\Rightarrow$ global optimality
- efficient solvers
- conceptual tools that generalize linear programming: duality, stopping conditions, ...

Local minima are global for convex functions

Theorem

If $x^\star$ is a local minimizer of a convex function f, then $x^\star$ is a global minimizer.

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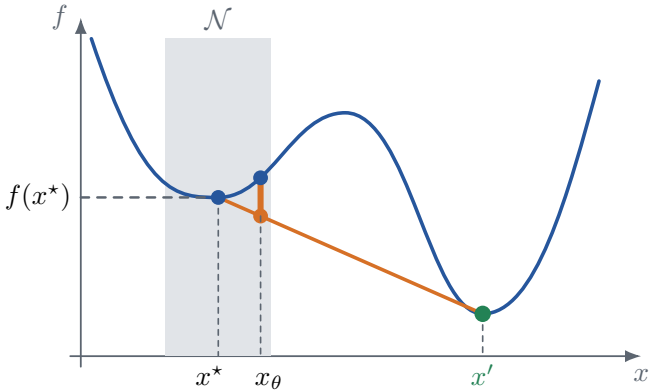
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proof? suppose by contradiction that another point x' is a global minimizer, with $f(x') < f(x^\star)$. draw the chord between x' and $x^\star$. since the chord lies above f, every convex combination $x = \theta x^\star + (1-\theta)x'$ of x' and $x^\star$ for $\theta \in (0,1)$ has a value $f(x) < f(x^\star)$. this is true even for $x \to x^\star$, contradicting our assumption that $x^\star$ is a local minimizer.

Local minima are global: picture



$f \geq f(x^\star)$ on $\mathcal{N}$, but the chord to x' dips below $f(x^\star)$ immediately. at $x_\theta \in \mathcal{N}$ the graph lies above the chord, so f cannot be convex.

Corollary

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If f is convex and differentiable and $\nabla f(x^\star) = 0$, then $x^\star$ is a global minimizer.

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Q: Is a global minimizer of a convex function always unique?
A: No. Picture.

Least squares: a global certificate

recall $f(x) = \frac{1}{2}\|Ax - b\|^2$. for every v,

$$v^T \nabla^2 f(x) v = v^T A^T A v = \|Av\|^2 \geq 0.$$

hence f is convex.

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$$A^T(Ax^\star - b) = 0$$

is a **global** minimizer.

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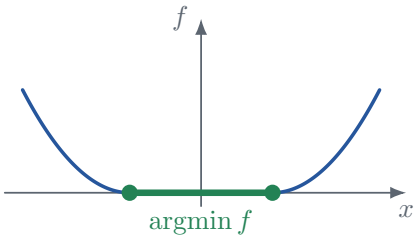
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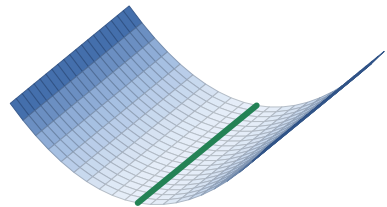
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is a **global** minimizer.

if A has full column rank, $A^T A$ is invertible, so the minimizer is unique.

Non-unique minimizers





left: a flat bottom. right: $\frac{1}{2}\|Ax - b\|^2$ with $A = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $b = 0$: A is not full column rank and the minimizers form a line.

Summary

after today, you should be able to:

- assess whether a point is a local or global minimizer
- state and apply first- and second-order optimality conditions
- define convex sets and functions
- prove convexity using different definitions and operations that preserve convexity
- state and apply Jensen's inequality
- compute subgradients of simple functions
- certify global optimality using gradients or subgradients